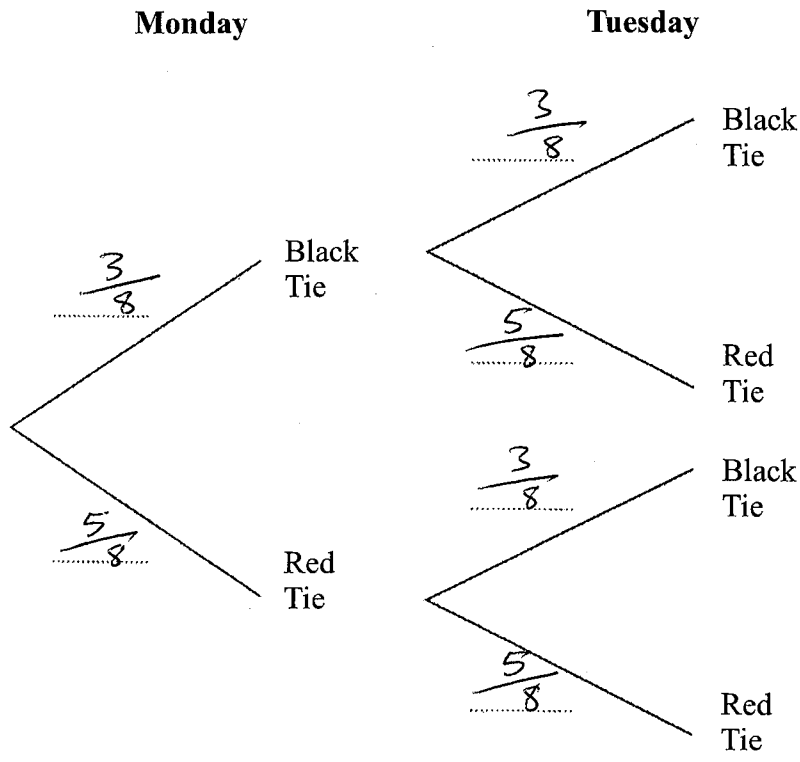
1 Each day Paul wears either a black tie or a red tie to work.

On any day the probability he wears a black tie is $\frac{3}{8}$

(a) Complete the probability tree diagram for Monday and Tuesday.



(2)

(b) Work out the probability Paul wears different coloured ties on Monday and Tuesday .

B R $\frac{3}{8} \times \frac{5}{8} = \frac{15}{64}$

R B $\frac{5}{8} \times \frac{3}{8} = \frac{15}{64}$

$\frac{15}{64} + \frac{15}{64} = \frac{30}{64}$

$\frac{30}{64}$

(2)

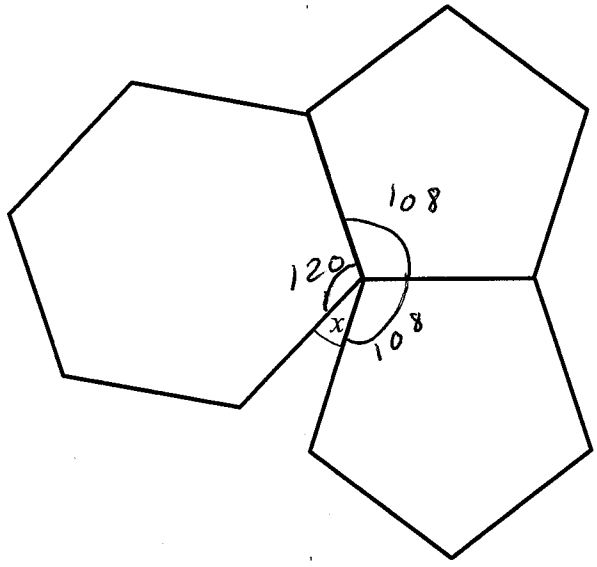
(Total for Question 1 is 4 marks)

2 A line passes through the point (0, -7).
The gradient of this line is 2.
Write down the equation of this line.

$y = 2x - 7$

(Total for Question 2 is 2 marks)

3



The diagram shows two regular pentagons and a regular hexagon meeting at a point.

Work out the size of the angle marked x.
You must show all your working.

$\frac{360}{5} = 72$ $\quad 180 - 72 = 108°$

$\frac{360}{6} = 60$ $\quad 180 - 60 = 120°$

$$\begin{array}{r} 120 \\ 108 \\ +\ 108 \\ \hline 336 \end{array}$$

$360 - 336 = 24°$

24 °

(Total for Question 3 is 3 marks)

4 Make b the subject of $a = \sqrt{\frac{b-7}{2}}$

$$a^2 = \frac{b-7}{2}$$

$$2a^2 = b - 7$$

$$2a^2 + 7 = b$$

$b = 2a^2 + 7$

(Total for Question 4 is 3 marks)

5 A is the point (7, –5) and B is the point (4, –1).

(a) Write down as a column vector $\overrightarrow{AB}$

$$\begin{pmatrix} 4 - 7 \\ -1 - -5 \end{pmatrix}$$

$\begin{pmatrix} -3 \\ 4 \end{pmatrix}$

(1)

C is the point (2, 5) and D is the point (–1, 9).

(b) Write down as a column vector $\overrightarrow{CD}$

$$\begin{pmatrix} -1 - 2 \\ 9 - 5 \end{pmatrix}$$

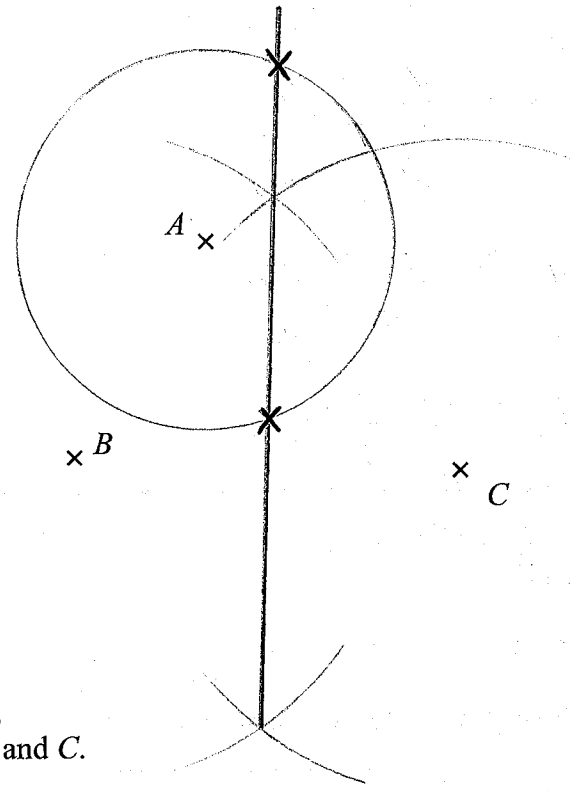
$\begin{pmatrix} -3 \\ 4 \end{pmatrix}$

(1)

(Total for Question 5 is 2 marks)

6 A, B and C are three points on a map.

1cm represents 100 metres.



Point P is 250 metres from A.
Point P is equidistant from B and C.

On the map, show the possible positions of P.

(Total for Question 6 is 3 marks)

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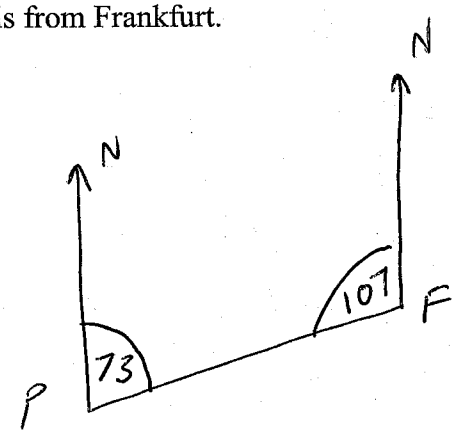
7 The bearing of Frankfurt from Paris is 073°

Find the bearing of Paris from Frankfurt.



360 − 107 = 253°

253 °

(Total for Question 7 is 2 marks)

8 The line AB passes through the points A(3, -2) and (6, k).

$x_1 \quad y_1 \quad x_2 \quad y_2$

The gradient of AB is 4.

Work out the value of k.

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$4 = \frac{k - -2}{6 - 3}$$

$$4 = \frac{k + 2}{3}$$

$$12 = k + 2$$

$$k = 10$$

$k =$ 10

(Total for Question 8 is 3 marks)

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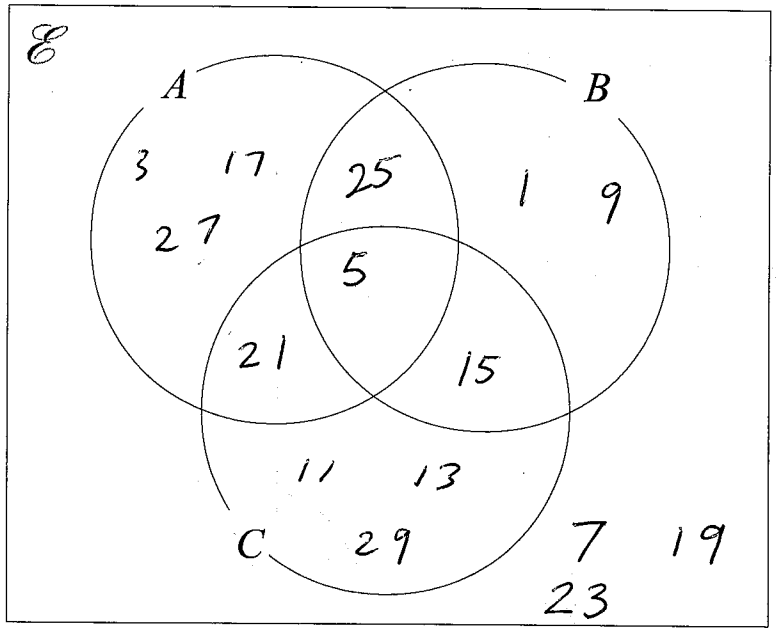
9 $\mathscr{E}$ = {odd numbers less than 30}
A = {~~3~~, ~~5~~, ~~17~~, ~~21~~, ~~25~~, ~~27~~}
B = {~~1~~, ~~5~~, ~~9~~, ~~15~~, ~~25~~}
C = {~~5~~, ~~11~~, ~~13~~, ~~15~~, ~~21~~, ~~29~~}

(a) Complete the Venn diagram to represent this information.



A number is chosen at random from $\mathscr{E}$.

(b) Find the probability that the number is a member of $(A \cap B)$.

$\frac{2}{15}$

(Total for Question 9 is 3 marks)